TugOS Build Gameplan

From Zero to Best-in-Class — The Methodical Route

Companion to the Vertical Architecture Blueprint v2 and Whitepaper v2

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This document is the execution layer on top of the TugOS Whitepaper v2 (the why and the what) and the Architecture Blueprint v2 (the how). It defines the end-state that 'best app in the field' actually means in measurable terms, sequences every build step against the blueprint's components, places a decision gate at the end of every phase, and maintains a risk register that pairs each failure mode with a prevention, an early-warning signal, and a contingency. It is a living document: every gate review updates it.

01 · Definition of Done: What 'Best in the Field' Means

'Best' is not a feeling; it is a set of conditions that can each be checked. TugOS is the best app in its field when, by roughly Month 36, all of the following are simultaneously true:

#	End-state outcome	How it is measured
O1	Crews choose it: captains and deckhands prefer TugOS to paper	Weekly active vessels / contracted vessels at 80%+; crew-side app sessions on a majority of active vessels; the '55-year-old captain' usability test passes on every release
O2	It is the Subchapter M system of record for its customers	Customers pass TPO audits using TugOS exports alone; documented cases of USCG boardings where the Sub M record book was produced from the app in under 60 seconds
O3	It pays for itself visibly	Recovered billable hours and faster invoicing per customer exceed subscription cost; median invoice-send time under 24h from job completion
O4	The data moat is real	1,000+ vessel-months of operational data; AI maintenance and fuel models measurably outperform baseline rules (tracked vs. holdout)
O5	The segment knows it	75+ paying companies; an annual benchmark report cited in trade press; a port-authority reference; AWO presence every year
O6	The business stands on its own	$150K+ MRR, under 5% monthly logo churn, gross margin consistent with SaaS norms, default-alive without Series A

North-star metric: Weekly Active Vessels (WAV). One number that forces everything else to be true — a vessel is 'active' only when its crew logged a job, fuel entry, or maintenance item that week. Revenue follows vessels; vessels follow crews; crews follow UX.

02 · Operating Principles (apply to every decision)

- **Evidence before code.** Every feature traces to an interview quote, a support ticket, or usage telemetry. The audit that produced the v2 docs exists because desk assumptions failed; the same discipline applies to ourselves.
- **Crew-first, office-second.** The buyer is the owner, but the adoption decision is made in the wheelhouse. Every feature ships only after a working captain has used it.
- **Compliance is the spine, not a module.** 46 CFR Subchapter M / TSMS record-keeping is designed into the data model from the first schema migration, not bolted on in Phase 2.
- **Offline is a feature of the foundation.** Sync correctness is tested from the first mobile build; a single beta data-loss event halts scaling until root-caused.

- **License hygiene by construction.** No copyleft or non-commercial code in the product path; a license check runs in CI. (This is how the CC BY-NC AIS toolbox error stays fixed.)
- **Autonomy is earned, not granted.** AI agents climb a ladder — log-only, then supervised, then autonomous-with-interrupt — and financial or compliance actions never skip rungs.
- **Gates, not vibes.** Each phase ends in a written gate review against pre-committed criteria. Passing is explicit; failing triggers the documented contingency, not improvisation.

03 · The Reference Architecture and Its Build Order

The Architecture Blueprint v2 defines the destination: three surfaces (UI, REST API, MCP), five MCP servers plus NotebookLM memory, six governed agents, seven scheduled automations, the Catalog of Authority, a dedicated Supabase project with per-tenant RLS, and offline refresh-token auth. The gameplan's job is sequencing — what gets built when, and why.

Blueprint component	Phase	Sequencing rationale
Genetic template deploy + dedicated Supabase project + core tug_ schema	0–1	Everything sits on it; isolation decisions are cheap now, expensive later
Surface 2 (Express REST API) + roles/auth	1	Every UI action is an API call; the API is the contract everything else builds against
Surface 1 (dispatch board UI)	1	The MVP workflow; the thing that replaces the radio log and phone calls
Mobile app core (offline job log)	1	Crew adoption is the wedge; offline correctness must mature for 12+ months before it matters at scale
Real-time layer — Supabase Realtime (NOT WebSockets on Vercel)	1	Verified: Vercel serverless cannot hold WebSocket connections; Supabase Realtime subscriptions deliver the live job board within the chosen stack; a Fly.io socket service is the fallback if Realtime limits bind
Billing engine + QuickBooks sync	2	Highest visible ROI; needs trustworthy job records as input, which Phase 1 produces
Sub M / TSMS compliance module	2	Highest retention lever; needs advisor + TPO validation loops that start in Phase 0
Fuel logging (manual first, Signal K optional)	2	Manual entry proves the habit; hardware integration only after software value is proven
Analytics / fleet P&L dashboard	2	Needs months of accumulated data to be non-empty; Vessel Finance patterns (money-as-cents, engines) port here
MCP surfaces (orchestration read-only, client portal)	2	AI-native surface starts read-only; write tools follow the autonomy ladder
Agents: Daily Briefing, Invoice (draft-only)	2	Lowest-risk agents first; both produce drafts a human approves
Agents: Fuel, Compliance, Dispatch, Fleet Health + Interrupt Protocol	3	Need 12+ months of baseline data to be better than rules; supervised before autonomous
Predictive maintenance models + benchmark report	3	The data moat monetized; requires vessel-month volume only Phase 2 retention can produce
Public API, marketplace (gated), international	3	Defensibility layers on top of an embedded product, never before it

04 · Phase 0 — Foundations (Weeks 1–4): Earn the Right to Build

Workstream A — Discovery (the main event)

- Recruit and retain the **maritime domain advisor** (retired port captain / ops manager) — they gate interview access, Sub M validation, and AWO credibility. Start here; week 1.
- Write the **interview instrument** around the four whitepaper hypotheses: H1 paper/Excel prevalence, H2 Sub M audit pain as purchase trigger, H3 willingness-to-pay ~$149/vessel below Helm, H4 spot-market liquidity. Add open workflow mapping: how a job arrives, who writes what, what a TPO audit looks like.
- Run **10–15 structured interviews** across operator sizes (solo, 3–10, 10–30 vessels, one port authority), recruited via the advisor, AWO Gulf Region, and port dispatcher relationships. Record, transcribe, and code against the hypotheses; every MVP feature must map to a quote.

Workstream B — Technical spike (parallel, timeboxed)

- Deploy the genetic template into a **dedicated Supabase project**; stand up core tug_ schema (vessels, jobs, crew, clients) with per-tenant RLS and write the first RLS isolation tests the same week.
- Prove the **real-time job board** path with Supabase Realtime subscriptions end-to-end (insert job, see board update); document the Fly.io fallback design.
- Prototype the **offline queue** on Expo: local store, append-only action log, sync replay, device-bound refresh token; kill-the-radio test on airplane mode.
- Evaluate **two commercial AIS APIs** on coverage, latency, and terms; wrap behind one adapter so the vendor is swappable. Bench-test Signal K on a single sensor kit but defer productization.
- Run the **license gate** across all planned dependencies (no NC/copyleft in product path; DocuSeal AGPL decision documented) and wire it into CI.

> **Gate 0 (end of Week 4).** PASS requires: advisor retained · 10+ interviews coded · H2 confirmed by a majority of mid-size operators (Sub M pain in their top 3) · H3 signal present (multiple operators say yes at ~$149/vessel) · RLS isolation tests green · offline prototype survives airplane-mode replay with zero loss. FAIL on H2/H3 → reposition before building (see Kill/Pivot, section 09). Do not start Phase 1 on a failed gate.

05 · Phase 1 — Trust on the Water (Months 1–6)

One workflow, nailed: digital job dispatch with crew notification and time-stamped logging. If a dispatcher can replace the radio log and phone calls in 90 days, everything else follows.

Build sequence

- M1–M2: Surface 2 API (jobs, vessels, crew, clients; role middleware) → dispatch board UI with Realtime updates → mobile job acceptance + status progression (en route / on scene / complete / clear) with offline queue.
- M2–M3: time-stamped job logging with photo capture; GPS breadcrumbs on active jobs; draft-invoice export (manual trigger, PDF via pdf-lib) — early because recovered billables are the first visible ROI.
- M3–M6: hardening loop with betas — sync telemetry (sync-failure rate, conflict rate, crash-free sessions) on a dashboard watched weekly; UX iteration with working captains aboard.

Beta program

- 5 Gulf Coast operators, comped 6 months, in exchange for weekly feedback calls, telemetry access, and a case study. Onboard in person; the advisor joins the first call with each.
- Instrumentation from day 1: every feature emits usage events; WAV is computed from week one.

> **Gate 1 (end of Month 6).** PASS requires: 5 active betas · 500+ jobs logged · 0 data-loss events · WAV at 70%+ of beta vessels · 2+ dispatchers state they would not go back to paper (recorded). FAIL on adoption → UX root-cause with captains before any new features. FAIL on data loss → freeze features, fix sync, re-run 30-day soak.

06 · Phase 2 — Become the Operator's OS (Months 6–18)

- **Sub M / TSMS module:** drills, inspections, maintenance logs, manning records, audit-readiness sweeps (the blueprint's subm-audit-readiness automation). Validate the record formats with the advisor AND a practicing TPO auditor before launch; pilot with one beta through a real audit cycle. This is the retention moat: years of compliance records that live nowhere else.
- **Billing engine:** hourly assist / bollard pull / charter / contract models; two-click invoice from completed job with GPS track and crew list attached; QuickBooks two-way sync; disputed-invoice workflow. Measure recovered billables per customer and publish the number to the customer monthly.
- **Captain app completion:** fuel log entry, maintenance reporting with photos, client signature capture (DocuSeal behind the adapter, pending license review), push notifications. 90/30/7 cert-expiry alerts to crew and office.
- **Analytics layer:** utilization, revenue per operating hour, fuel trend, top clients — porting Vessel Finance's proven patterns (integer-cents money, profitability/forecast engines).
- **First agents (draft-only):** Daily Briefing (6am compile) and Invoice (draft within 1 hour of job completion, human approves). Both run weeks in log-only mode first; their outputs are compared to human baselines before anyone sees them.
- **Trust infrastructure:** SOC 2 Type I via Vanta/Drata (made credible by the dedicated project isolation); third-party pen test focused on cross-tenant access before the first port authority conversation; public status page.
- **Port authority reference:** target the 3rd–4th largest Gulf port (they move faster); the pen test, SOC 2, and the advisor open the door.

> **Gate 2 (Month 18).** PASS requires: 20+ paying companies · $40K+ MRR · <5% monthly logo churn · 200+ crew profiles · 1+ customer passed a TPO audit using TugOS records · SOC 2 Type I issued · invoice agent draft-acceptance above 80%. FAIL on churn → stop acquisition spend, run win-back interviews, fix the top-2 cancellation causes before resuming. FAIL on audit validation → compliance module stays 'beta'-labeled until a real audit passes.

07 · Phase 3 — Win the Category (Months 18–30)

- **Agent suite to full strength** under the Interrupt Protocol: Fuel (auto-limited under $2K), Compliance (supervised; Sub M completeness), Dispatch (supervised recommendations), Fleet Health (autonomous scoring). Each climbs log-only → supervised → autonomous with pre-committed promotion criteria (accuracy vs. human baseline over a defined window) and automatic demotion on error.
- **Predictive maintenance + fuel anomaly models** trained on accumulated fleet data; performance tracked against a holdout and against the simple-rules baseline — the moat claim must be measured, not asserted.
- **Annual 'State of Tugboat Operations' benchmark report** from anonymized aggregate data; launch at AWO/Workboat; this is the inbound engine and the authority position.
- **Public API + integration partners:** payroll, insurance cert validation, port systems. Each integration deepens switching costs.
- **Spot Job Board marketplace — built ONLY if hypothesis H4 tested true** in Phases 0–2 evidence (operators repeatedly asking for spot capacity matching). If liquidity is absent, the network effect comes from benchmark data instead; do not build a ghost-town marketplace.
- **International expansion** (IMO/STCW frameworks travel; multi-currency; Spanish, Bahasa) and **Series A** from strength: 75+ customers, $150K+ MRR, measured data moat — or skip the raise if default-alive economics make it optional.

> **Gate 3 (Month 30).** PASS requires: 75+ paying · $150K+ MRR · 1,000+ vessel-months · agent suite at target autonomy with zero uncontained high-stakes errors · benchmark report published and cited. FAIL on agent safety → demote autonomy and extend supervised period; the Interrupt Protocol's audit trail is the evidence

base either way.

08 · Phase 4 — Best in Class as a Posture (Months 30–36+)

By Month 30 the remaining work is organizational, not architectural: the data moat compounds (every vessel-month deepens the deficit later entrants face); crew UX is re-validated on every release against the wheelhouse test; compliance depth means customers run actual USCG boardings from the app; community trust is renewed annually at AWO and Workboat with real customer stories; the public roadmap, quarterly customer advisory board, and 24-hour bug SLA keep the feedback flywheel spinning. Best-in-class is held, not reached: section 01's outcomes O1–O6 are re-measured quarterly forever.

09 · Risk Register — What Could Go Wrong, and the Countermeasure

Each risk carries a prevention (built in advance), an early-warning signal (watched on a dashboard or in gate reviews), and a contingency (pre-decided response). The register is reviewed monthly and at every gate.

A · Market & competitive

Risk	Prevention	Early warning	Contingency
Helm CONNECT moves down-market with an SMB tier	Speed + self-serve + crew UX as structural advantages an enterprise vendor is slow to copy; transparent pricing as the anchor	Helm pricing/packaging announcements; losses in deals citing Helm	Compete on adoption speed and wheelhouse UX; never on feature count
Discovery invalidates H1/H3 (market is more digital or poorer than assumed)	Phase 0 gate exists precisely for this; hypotheses pre-written	Interview coding shows <40% hypothesis confirmation	Reposition (different segment, price, or wedge workflow) before any build spend
Marketplace assumption fails (H4)	Marketplace is gated, not scheduled; benchmark-data network effect is the alternative	No organic spot-capacity requests from customers by Phase 2	Ship benchmark reports as the network layer; revisit marketplace yearly
A VC-funded copycat outspends marketing	Compliance records + data history = switching costs that ads cannot buy	Funded entrant announcements; CAC inflation	Double down on retention levers (Sub M records, analytics); publish the benchmark report

B · Adoption & product

Risk	Prevention	Early warning	Contingency
Crews reject the app (the classic maritime-software death)	Wheelhouse test every release; offline-first; captain co-design in betas; the advisor's veto on UX	WAV below 70% of contracted vessels; crew sessions concentrated in office roles	Feature freeze; field visits; fix the top-3 crew complaints before anything else
Beta data loss destroys trust	Append-only sync log; soak tests; sync telemetry watched weekly; 0-loss gate	Any sync-failure or conflict-rate uptick	Halt scaling; root-cause; 30-day clean soak before resuming
Feature bloat dilutes the MVP	Every feature maps to an interview quote; one-workflow MVP discipline	Backlog items without quotes attached	Cut scope at gate reviews; the gate criteria do not include feature count
Free tier attracts support load, not conversions	Deckhand tier capped (1 vessel, 30-day history); self-serve onboarding	Support tickets per free account; free-to-paid conversion under 3%	Tighten tier limits; add in-product upgrade prompts; never staff support for free tier

C · Technical & architecture

Risk	Prevention	Early warning	Contingency
Real-time job board fails on serverless (verified: Vercel cannot host WebSockets)	Supabase Realtime subscriptions from day 1; adapter so transport is swappable	Realtime latency/connection limits in load tests	Dedicated Fly.io socket service (design documented in Phase 0)
Offline sync conflicts corrupt records	Append-only action log + server-side replay with deterministic conflict rules; property-based sync tests	Conflict-rate telemetry; support tickets mentioning 'missing' entries	Freeze, reproduce, fix, soak; compensating audit log preserves every raw action

Risk	Prevention	Early warning	Contingency
Cross-tenant data leak (RLS misconfiguration)	Dedicated project; RLS tests in CI per table; tenant-scoped connections; pen test before port-authority deals	Any RLS test regression; pen-test findings	Disclose per SOC 2 obligations; incident runbook written in Phase 2
AIS vendor cost or terms shift	Two vendors evaluated; one adapter interface; usage caps and alerts	Per-vessel AIS cost trending above model	Swap vendor behind the adapter; degrade to lower polling before degrading UX
Signal K hardware variance burns integration time	Manual fuel entry first; IoT optional per vessel; supported-hardware list kept short	Integration hours per vessel rising	Pause hardware onboarding; certify one kit; revisit quarterly
Genetic-template drift (dedicated project diverges from OSL core)	Template updates pulled on a cadence; deviations documented in the blueprint	Merge pain on template updates	Quarterly reconciliation sprint; never fork silently

D · Compliance & legal

Risk	Prevention	Early warning	Contingency
Sub M records implemented wrong (worst case: a customer fails an audit using our exports)	Advisor + practicing TPO auditor validate formats before launch; pilot through a real audit; module labeled beta until one passes	TPO feedback on pilot; USCG guidance changes	Immediate correction release; customer notification; regulatory-watch automation
License violation ships (the CC BY-NC near-miss, repeated)	License gate in CI; no NC/copyleft in product path; AGPL components isolated and reviewed	CI license-gate failures; dependency-bump alerts	Remove/replace component; legal review; document in the register
Crew PII mishandled	PII changes route through the Interrupt Protocol; data minimization; SOC 2 controls	Access-log anomalies	Incident runbook; disclosure per obligations
Advisor key-person dependency	Build an advisory board of 3+ by Phase 2 (the gameplan's customer advisory board doubles here)	Single advisor on every critical path	Retain a second advisor; document advisor knowledge in the wiki

E · AI & agents

Risk	Prevention	Early warning	Contingency
An autonomous agent takes a wrong high-stakes action	Autonomy ladder (log-only → supervised → autonomous); Interrupt Protocol on financial/compliance/PII actions; MLflow audit trail	Draft-acceptance rate falling; any interrupt override	Automatic demotion one rung; incident review; promotion criteria re-run from zero
Beacon assistant hallucinates a regulatory answer	RAG over the Catalog of Authority with mandatory citations; advisor spot-audits; 'consult your TPO' framing on compliance answers	Citation-missing rate; user corrections	Constrain to retrieval-only answers for regulatory queries
Model/API cost outgrows unit economics	Per-tenant AI cost telemetry from first agent; caching; small-model routing	AI cost per vessel-month trending up	Batch schedules; cheaper models for routine drafts; reprice Enterprise tier

F · Business & execution

Risk	Prevention	Early warning	Contingency
Runway burns before Gate 2 economics arrive	Phase budgets set at Gate 0; default-alive plan; comped betas capped at 5	Burn vs. budget monthly; gate slippage	Cut scope to the retention core (dispatch + Sub M + billing); extend timeline, not burn
Churn from support gaps as customer count grows	24h bug SLA for paying customers; public roadmap; quarterly advisory board	Time-to-first-response; churn interviews citing support	Hire support before sales; the gameplan staffs retention before acquisition
Pricing set wrong (too low to sustain, too high to land)	H3 tested in Phase 0; recovered-billables ROI published per customer	Win/loss notes; discount frequency	Reprice with grandfathering; value-anchor on recovered revenue, not features
Series A dependence forces bad terms	Default-alive as the planning baseline; raise from strength or not at all	Plan requires capital before Gate 2 passes	Slow hiring; focus on Gulf density over geographic spread

10 · Kill / Pivot Criteria (decided now, while heads are cool)

- **Reposition** if Gate 0 fails on H2/H3: the segment, price point, or wedge workflow changes before MVP code is written.
- **Halt scaling** on any beta data-loss event or cross-tenant finding: features freeze until a clean 30-day soak passes.
- **Stop acquisition** if monthly logo churn exceeds 8% for two consecutive months: retention is fixed before another dollar of marketing.
- **Do not build the marketplace** unless H4 evidence exists by Gate 2; revisit annually.
- **Re-plan the venture** if Gate 1 passes but Gate 2's paying-customer criterion misses by more than 50% at Month 18: the honest options (niche lifestyle business, acquihire conversations, or a different vertical on the same template) get a written decision memo.

11 · Measurement System

Layer	Metrics	Cadence
North star	Weekly Active Vessels (WAV) and WAV / contracted vessels	Weekly
Adoption	Crew-side sessions per vessel; jobs logged; time-to-first-job for new customers; '55-year-old captain' test pass rate	Weekly
Reliability	Sync-failure rate; conflict rate; crash-free sessions; data-loss events (target: zero, forever)	Continuous dashboard
Economics	MRR; logo + revenue churn; CAC by channel; recovered billables per customer; AI cost per vessel-month	Monthly
Compliance value	Customers through TPO audits on TugOS records; inspection-report generation time	Per event + quarterly
AI quality	Draft-acceptance rates; model-vs-baseline lift on holdout; interrupt overrides	Weekly once agents ship
Risk	Register review; early-warning signal scan	Monthly + every gate

12 · Governance & Cadence

- **Weekly:** beta/customer feedback call; reliability dashboard review; WAV review.
- **Monthly:** risk-register review; burn vs. phase budget; metrics pack.

- **Per gate:** written gate review against the pre-committed criteria in this document; the gameplan is amended in writing, never informally.
- **Quarterly (from Phase 2):** customer advisory board (5–8 operators); roadmap publication; O1–O6 re-measurement.
- **Always:** every feature ships with telemetry; every incident gets a written post-mortem that updates the risk register; the wiki and this document stay in sync (Limitless Stack session discipline).

13 · Audit Record (audit-before-claim)

This plan distinguishes verified facts from working assumptions. Verified this session (2026-06-04, web-checked): Helm CONNECT's market position and Sub M marketing; 46 CFR Subchapter M as the central towing compliance regime; IMO CII applying at 5,000 GT and above; the MarineTraffic AIS Toolbox's CC BY-NC-SA license; work-hour rules at 46 U.S.C. 8904 / 46 CFR 15.1111; Vercel serverless functions' inability to host WebSocket connections; market size estimates ranging $1.2–2.4B at 10–17.5% CAGR across research firms.

Working assumptions, to be validated and never treated as facts: H1–H4 (legacy prevalence, Sub M purchase trigger, willingness-to-pay, marketplace liquidity); all pricing; all phase targets and gate thresholds (chosen judgments, not derived constants); agent performance expectations; Supabase Realtime fitness at production scale (proven only to spike level in Phase 0). Every gate review re-tests the assumptions its phase depended on.